



UNITED INTERNATIONAL UNIVERSITY (UIU)
DIGITAL SIGNAL PROCESSING LABORATORY
COURSE CODE EEE 3310

Lab Report : 06

DFT Computation Using FFT Algorithm in MATLAB

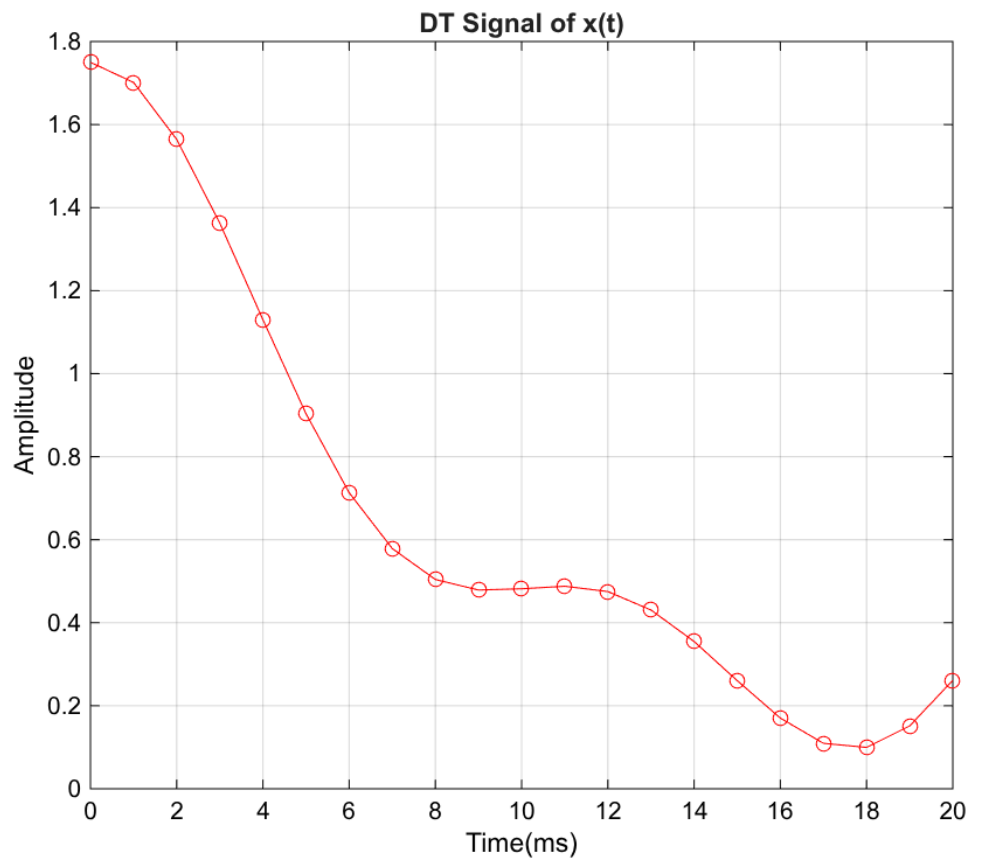
Group No: 02

Group Members:

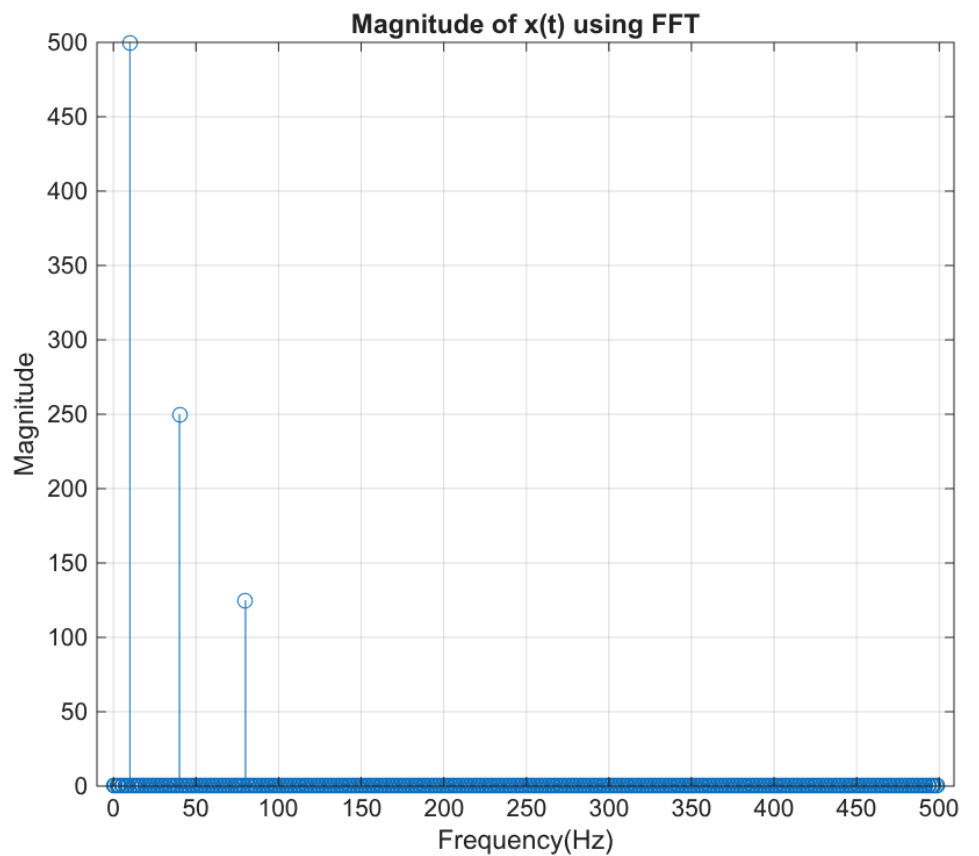
0212330092 Md. Hussain Shahriar Rokon
021213012 Rifa Akter
0212320035 Nowshin Tabassum Rihun

Answer: 6.2 Exercise (HW6.01)

(i) Output Figure



(ii) Output Figure



Matlab Code:

```
clc; clear all; close all;  
Fs = 1000;  
Ts = 1/Fs;
```

% Part (i)

```
t = 0:Ts:0.02; %time array for 20 ms  
x = cos(20*pi*t) + 0.5*cos(80*pi*t) + 0.25*cos(160*pi*t);
```

```
figure  
plot(t*1000,x,'r-o');  
title('DT Signal of x(t)');  
xlabel('Time(ms)');  
ylabel('Amplitude');  
grid on;
```

% Part(ii)

```
t_1 = 0:Ts:1-Ts; % t = 1 second for better resolution  
x = cos(20*pi*t_1) + 0.5*cos(80*pi*t_1) + 0.25*cos(160*pi*t_1);  
N = length(x); % FFT points  
X = fft(x, N);  
k = 0:N-1; % frequency index  
F = (Fs/N)*k; % frequency axis in Hz
```

```
figure  
stem(F(1:N/2), abs(X(1:N/2)));  
% 1:N/2 for first half points, and this first half will repeat  
% abs for the magnitude only  
title('Magnitude of x(t) using FFT');  
xlabel('Frequency(Hz)'); ylabel('Magnitude');  
grid on;
```